

Table 1- Summary of genome features derived from Pharokka/Phold/Phyteny re-annotation of Microviridae reference genomes and patient phages

Sample ID	Genome Length (bp)	#CDS	#Hypothetical Proteins	#Annotated Proteins	% Hypothetical Proteins	GC%	AT%
<i>Escherichia_phage_EC6098</i>	4526	7	2	5	28.6	48.4	51.6
<i>Phage_phiMH2K</i>	4594	9	4	5	44.4	46.1	53.9
<i>Spiroplasma_Phage_4</i>	4421	17	10	7	58.8	32.1	67.9
<i>phage_comp_7804_cycle_1</i>	5832	12	6	6	50	45.4	54.6
<i>phage_comp_7963_cycle_1</i>	5978	13	6	7	46.2	43.9	56.1
<i>phage_comp_7997_cycle_1</i>	5749	12	6	6	50	43.3	56.7
<i>phage_comp_7941_cycle_1</i>	6062	15	9	6	60	44.6	55.4

Spiroplasma phage 4 reference genome contains only a single annotated CDS in its original GenBank record (capsid protein). Re-annotation using Pharokka/Phold/Phyteny predicted 17 CDS, 7 being known of which 4 code for major head proteins and 3 code for replication initiation proteins.

The Escherichia phage EC609 reference genome contains 6 annotated CDS in its original GenBank record. Re-annotation using Pharokka/Phold/Phyteny predicted 7 CDS, but the additional is hypothetical.

The Phage phiMH2K reference genome contains 11 annotated CDS in its original GenBank record. Re-annotation using Pharokka/Phold/Phyteny predicted 12, reorientating the genome actually annotated more than what is annotated in the GenBank Record, with fewer hypothetical proteins comprising the CDS, which is different from the other two closely related microviridae that re-annotation introduced more CDS which code for the protein products that are hypothetical.

Table 2- Comparative functional annotation profile of Microviridae patient genomes and reference genomes

Protein Function	phage_comp_7804_cycle_1	phage_comp_7963_cycle_1	phage_comp_7997_cycle_1	phage_comp_7941_cycle_1	Escherichia _phage_EC609	Phage_phiMH2K	Spiroplasma_Phage_4
Hypothetical protein	6	6	6	9	2	4	10
virion structural protein	1	1	1	1	1	1	
replication initiation protein	1	1	1	1	1	1	3
pilot protein for DNA ejection	1	2	1	1	1		
major head protein	1	1	1	1	1	1	4
endolysin	1	1	1	1			
DNA binding protein	1	1	1	1	1	1	
minor tail protein						1	

Notes:

The four patient-derived Microviridae genomes encode a conserved functional gene set, suggesting they are closely related and share similar genome organisation, which we already knew through the MashTree/IToL pipeline.

Endolysin = a phage-encoded enzyme that **breaks bacterial cell walls** to allow phage release.

All patient-derived phages encoded an endolysin, indicating conserved lysis machinery and suggesting these phages **may** contribute to bacterial turnover in the gut environment.

Endolysins as antimicrobials: The consistent presence of an endolysin gene across patient phages highlights a potential therapeutic relevance, as phage-derived endolysins are increasingly explored as targeted antimicrobial agents.

But why does SP4 have similar protein outputs, but shows no shared sequences with the other two NCBI microviridae?

SP4 *must* have equivalents.

But:

- They may be **highly diverged**
- They may share **no detectable sequence similarity**

Yes, SP4 will have similar *types* of genes — but not necessarily similar *sequences*.

Our output showed:

- EC6098 ↔ phiMH2K → 30–40% identity matches
- SP4 ↔ EC6098 or phiMH2K → no reported matches

That suggests:

phiMH2K is evolutionarily closer to EC6098

SP4 is more distantly related

Table 3- Functional Gene Composition (% of total CDS)

Functional Category	<i>phage_com p_7804_cyle_1 (%)</i>	<i>phage_com mp_7963_cycle_1 (%)</i>	<i>phage_com mp_7997_cycle_1 (%)</i>	<i>phage_com mp_7941_cycle_1 (%)</i>	<i>Escherichia_phag e_EC609 (%)</i>	<i>Phage_phiM H2K (%)</i>	<i>Spiroplasma_Phag e_4 (%)</i>
<i>Hypothetical protein</i>	50	46.2	50	60	28.6	44.4	58.8
<i>virion structural protein</i>	8.3	7.7	8.3	6.7	14.3	11.1	0
<i>replication initiation protein</i>	8.3	7.7	8.3	6.7	14.3	11.1	17.6
<i>pilot protein for DNA ejection</i>	8.3	15.4	8.3	6.7	14.3	0	0
<i>major head protein</i>	8.3	7.7	8.3	6.7	14.3	11.1	23.5
<i>endolysin</i>	8.3	7.7	8.3	6.7	0	0	0
<i>DNA binding protein</i>	8.3	7.7	8.3	6.7	14.3	11.1	0
<i>minor tail protein</i>	0	0	0	0	0	11.1	0

At present for the patient phages approximately half of the coding codes for proteins whose function is yet to be identified. However, of the four patient phages, there are similarities in the types of proteins output as well as the percentage of the genome that codes for these protein sub-categories. The only notable difference is the % of the CDS that encodes for the pilot protein for DNA ejection, of which phage 7963 has almost double the % that the other three phages accommodate in their CDS regions of their genomes.

Comparing the three annotated NCBI phages, Spiroplasma Phage is less similar than Escherichia Phage and Phage_phiMH2k, the latter two are more closely related when concerning the percentage of protein-subcategories in the CDS.

When comparing the NCBI phages to the patient phages, the Spiroplasma phage shows less similarity than the other two NCBI phages. The Escherichia phage and the Phage_phiMH2K are similar in all protein output categories to the patient phages, with the exception of pilot protein for DNA ejection for the Phage_phiMH2K which has a genome that does not encode for that protein output, and the Escherichia phage with a minor tail protein output, which the CDS is not found in any of the patient phages or the other two NCBI phages. Endolysin is neither a protein output for either Escherichia phage or Phage_phiMH2K, but is for the four patient phages which possess the coding sequence. This suggests that while the overall genome structure is similar, the way these phages lyse bacterial cells may differ.. It is noted that there is a greater proportion of the CDs region of the genome that encodes for the functional proteins in the NCBI phages, and less of the CDS codes for hypothetical proteins.

Table 4- Gene conservation and amino acid identity between patient-derived phages and *Escherichia phage EC6098*.

- Identity values represent amino acid sequence identity from Clinker alignments.

PATIENT PHAGE	CLOSEST NCBI REFERENCE	# CONSERVED GENE REGIONS	MEAN IDENTITY %	MAX IDENTITY %
PHAGE_COMP_7804_CYCLE_1	Escherichia_phage_EC6098	2	41	45
PHAGE_COMP_7963_CYCLE_1	Escherichia_phage_EC6098	3	37	40
PHAGE_COMP_7997_CYCLE_1	Escherichia_phage_EC6098	3	34	37
PHAGE_COMP_7941_CYCLE_1	Escherichia_phage_EC6098	3	35	42

- Conserved regions defined as gene clusters meeting Clinker homology thresholds.
- Mean identity calculated across conserved gene clusters per phage.

Notes:

Despite limited gene conservation (2–3 shared clusters), the conserved regions exhibited moderate amino acid similarity (mean 34–41%), suggesting structural conservation within a divergent lineage. Notably, phage_comp_7804_cycle_1 displayed the highest mean (41%) and maximum (45%) identity, indicating comparatively stronger evolutionary constraint among its shared genes.

Max identity → strongest conserved functional anchor

- These are not identical phages.
- They are related but evolutionarily diverged.
- Genome structure is conserved more than sequence identity.